ModelIt!

Teacher's Guide

The Heat Island Effect

Modeling Urban Temperature Amplification, Health Impacts, and Green Infrastructure Solutions

Quick Reference

Grade Level	Grade 11L2
NGSS Standard	HS-ESS3-4, HS-ETS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-4, HS-ETS1-1

Evaluate or refine a technological solution that reduces impacts of human activities on natural systems. Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct evidence-based explanations of urban heat island formation and design green infrastructure solutions informed by computational model data on surface albedo, evapotranspiration, and heat-health relationships.
Disciplinary Core Idea	**ESS3.C Human Impacts on Earth Systems / ETS1.A Defining and Delimiting Engineering Problems** — Students model how urban development amplifies local temperatures with cascading health consequences, then define the engineering problem of reducing heat exposure while accounting for equity, economics, and community impact constraints.
Crosscutting Concept	**Cause and Effect / Systems and System Models** — Students trace causal chains from urban land cover decisions through thermodynamic processes to health outcomes, modeling the city as a system where material choices, energy flows, and human vulnerability interact to produce unequal heat exposure.

CAST Testing Alignment

MULTIPLE CHOICE:

A city's thermal survey reveals that the historically redlined neighborhood with 5% tree canopy and 90% impervious surface is 12°F hotter than the affluent neighborhood 2 miles away with 35% tree canopy and 40% impervious surface. Climate change projections add 4°F to baseline temperatures by 2050. Based on your model, which statement BEST describes the combined effect on the underserved neighborhood?

CONSTRUCTED RESPONSE:

Using your model, explain how a decision made in the 1930s — redlining a neighborhood, denying investment, and allowing tree canopy to disappear — can result in heat-related deaths in that same neighborhood 90 years later. Reference Urban Land Cover, albedo, evapotranspiration, thermal mass, and the concept of environmental justice in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The urban heat island effect is one of the most measurable and consequential ways that human-built environments alter local climate. Understanding this phenomenon requires modeling the intersection of thermodynamics, urban planning, public health, and environmental justice to reveal why the places where most humans live are also the places most dangerously amplifying the heat that climate change is already delivering.

The Physics of Urban Heating

Urban heat islands form through three primary mechanisms. First, albedo reduction: natural surfaces like vegetation and soil reflect 20-30% of incoming solar radiation, while urban materials absorb 80-95% — dark asphalt (albedo 0.05-0.10) converts nearly all sunlight to stored thermal energy. Second, thermal mass storage: concrete, brick, and asphalt have high thermal mass, absorbing heat throughout the day and releasing it slowly at night — this is why cities remain hot after sunset while rural areas cool rapidly, creating dangerous nighttime heat that prevents the human body from physiological recovery. Third, anthropogenic waste heat: vehicles, air conditioners, industrial processes, and electrical equipment generate heat that directly warms the urban atmosphere — in dense city cores, waste heat can add 2-4°F to ambient temperature. The combined effect creates temperature differentials of 5-12°F during the day and up to 22°F at night compared to surrounding rural areas.

The Environmental Justice Dimension

Urban heat is not distributed equally. Research consistently shows that the hottest neighborhoods in American cities are those that were redlined in the 1930s — designated as 'hazardous' by federal housing appraisers, denied investment, and populated predominantly by Black and Brown residents. These neighborhoods received fewer parks, fewer trees, more highways, more industrial facilities, and more impervious surface. Today, formerly redlined neighborhoods are an average of 5°F hotter than non-redlined neighborhoods in the same city, and some show differentials exceeding 12°F. The consequences are measured in lives: heat-related mortality rates are 2-4 times higher in low-canopy neighborhoods. Residents in these areas are also more likely to be outdoor workers, less likely to have air conditioning, and more likely to have pre-existing health conditions that heat exacerbates. The urban heat island is a physical manifestation of structural racism — past decisions about where to invest literally determine who is safe from heat today.

Health Impacts and Thresholds

Heat is the deadliest weather phenomenon in the United States, causing more deaths annually than hurricanes, floods, tornadoes, and lightning combined. Heat-related illness follows a nonlinear threshold pattern: the human body can thermoregulate effectively up to certain temperature-humidity combinations (wet-bulb temperatures below 35°C/95°F), but above these thresholds, physiological cooling mechanisms fail and core body temperature rises regardless of hydration or shade. Urban heat islands push already-hot cities past these thresholds for more hours per day and more days per year than rural areas experience. Climate change is projected to increase the number of dangerous heat days in major U.S. cities by 2-5 times by 2050. Nighttime heat is particularly dangerous because the body requires overnight cooling to recover from daytime heat stress — urban thermal mass prevents this recovery in ways that rural environments do not.

Green Infrastructure Solutions

Cooling cities requires a portfolio of complementary strategies. Tree canopy expansion is the most effective single intervention: each mature tree provides evapotranspiration cooling equivalent to 2-5 window air conditioning units, and increasing canopy from 10% to 30% can reduce neighborhood temperatures by 5-10°F. However, trees require 15-20 years to reach full cooling capacity, demanding long-term planning. Cool roofs — highly reflective surface coatings with albedo of 0.6-0.9 compared to conventional roofs at 0.1-0.2 — provide immediate temperature reduction of 2-5°F and cut building cooling costs by 10-30%. Permeable pavement allows water infiltration that provides evaporative cooling while also managing stormwater. Urban parks and green corridors create cool zones that extend 200-500 meters into surrounding neighborhoods. The most effective strategies combine all of these, prioritizing the hottest and most vulnerable neighborhoods. Critically, cooling investments

must include anti-displacement protections — research shows that greening historically underserved neighborhoods increases property values and can trigger gentrification that displaces the very residents the cooling was designed to protect.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify six components of the urban heat island system: Urban Land Cover Composition, Surface Heat Absorption Rate, Waste Heat Generation, Evapotranspiration Cooling Rate, Ambient Air Temperature, and Heat-Related Health Impact — distinguishing human-controlled land use decisions from the thermodynamic and biological processes that determine temperature distribution and health outcomes.

E **Establish Relationships**

Students determine that Urban Land Cover Composition is the primary driver: replacing vegetation with impervious surfaces increases Surface Heat Absorption Rate while reducing Evapotranspiration Cooling Rate. Waste Heat Generation adds anthropogenic warming. The balance of these three factors determines Ambient Air Temperature, which drives Heat-Related Health Impact nonlinearly above critical thresholds.

V **Visualize & Model**

Students build a computational model showing the energy balance of urban versus vegetated surfaces — visualizing how albedo, thermal mass, waste heat, and evapotranspiration interact to create temperature differentials across neighborhoods.

E **Evaluate Outcomes**

Students run current baseline, climate-amplified, and green infrastructure scenarios to quantify the heat differential between high-canopy and low-canopy neighborhoods, project dangerous heat days under climate change, and measure the cooling effectiveness of different intervention strategies.

R **Revise & Extend**

Students add Air Quality Degradation, Energy Demand Feedback, or Stormwater Runoff Intensity to explore how urban heat creates compound health hazards, drives self-reinforcing cooling demand cycles, and connects to flood vulnerability through shared impervious surface drivers.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.

At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Two neighborhoods. Same city. Same day. But one is 12 degrees hotter than the other. That difference isn't weather — it's a decision. And it was made decades ago."*

Watch For

Show an infrared thermal image of a city where the temperature differences between neighborhoods are visible in color gradients. Overlay a historic redlining map of the same city. Let students see the correlation. Ask: Why do the hottest neighborhoods match the redlined areas almost perfectly?

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling why cities are heat amplifiers — and why the people most affected by that heat are the same people who have been most disinvested in for a century."*

Watch For

Have students read objectives. Pre-teach albedo with a simple demonstration: hold a black cloth and a white cloth under a desk lamp for 60 seconds, then feel the temperature difference. Connect to surfaces: which is the parking lot and which is the park? Quick-write: Who decides where trees get planted in a city?

Slide 3: Big Question

3 min

Say: *"Heat kills more Americans than hurricanes, floods, and tornadoes combined. And within the same city, your risk of dying from heat can vary by 400% depending on your zip code. That's not nature — that's design."*

Watch For

Present the statistic: heat is the #1 weather killer in the U.S. Then show the within-city variation in heat mortality by neighborhood. Have students calculate: if heat-related ER visits are 4x higher in the hottest neighborhood, and that neighborhood has 5% tree canopy versus 30% citywide, what's the cost of missing trees measured in lives?

Slide 4: LEVER Framework

2 min

Say: *"We're modeling a system where every surface tells a story. Concrete, asphalt, glass — they're all absorbing heat and releasing it back into the air you breathe. And a tree is a better air conditioner than any machine humans have built."*

Watch For

Preview each LEVER step. Introduce the energy balance concept: every surface in a city either absorbs solar energy (heats up) or reflects/evaporates it (stays cool). The ratio of absorbing to cooling surfaces determines the temperature you feel. Draw it: sun energy in → surface decision → heat or cool.

Slide 5: Activity 1: Components

8 min

Say: *"One human decision — what we cover the ground with. Five system responses. But the responses are not equal for everyone in the city."*

Watch For

Guide sorting of external vs. internal components. Only Urban Land Cover Composition is external — it is determined by planning decisions, zoning codes, and investment patterns. Discuss why Surface Heat Absorption is internal: the physics of albedo and thermal mass operate automatically once the surface material is chosen. Connect to environmental justice: who made the land cover decisions, and who lives with the consequences?

Slide 6: Activity 2: Connections

8 min

Say: *"Follow the energy: sunlight hits the surface. Dark surfaces absorb it. No trees means no evaporative cooling. Air conditioning pushes heat outside. Temperature rises. People get sick. And it all started with a surface — and a decision about what surface to build."*

Watch For

Help students map the energy balance. Identify the air conditioning feedback loop: high temperatures → more AC use → more waste heat → higher temperatures → more AC use. Find the equity dimension: map component values for a high-canopy wealthy neighborhood versus a low-canopy underserved neighborhood. Same sun, same city — different outcomes.

Slide 7: Activity 3: Simulation

10 min

Say: *"Three futures for this city: do nothing, add climate change, or invest in green infrastructure where it's needed most. The temperature numbers matter. But the health numbers — and who they fall on — matter more."*

Watch For

Students predict before running simulations. Key focus: the equity gap. In the baseline scenario, track the temperature and health impact difference between the 'greenest' and 'least green' neighborhoods. In the climate-amplified scenario, observe how many additional dangerous heat days the underserved neighborhood faces. In the green infrastructure scenario, evaluate whether targeted investment can close the gap.

Slide 8: Discoveries

5 min

Say: *"Your model just revealed that the urban heat island is a map of decisions — decisions about where to plant trees, where to build highways, where to invest, and where not to. The physics is simple. The injustice is not."*

Watch For

Lead discussion connecting urban heat to environmental justice. Show the redlining-to-heat-mortality pipeline. Ask: If tree canopy in the hottest neighborhood was raised to match the city average, how many lives would be saved? What has prevented that investment? Discuss gentrification risk: how do we cool neighborhoods without pricing out the people who live there?

Slide 9: STEM Challenge

12 min

Say: *"A neighborhood where heat-related ER visits are four times the city average. Five percent tree canopy. Eighty-five percent concrete. Your job: design the cooling plan that saves lives — without displacing the people those lives belong to."*

Watch For

Groups design comprehensive neighborhood cooling plans including short-term interventions (shade structures, cool roofs, misting stations), long-term investments (tree planting with species selection, park creation, permeable pavement), community engagement strategy, anti-displacement protections, and 15-year temperature reduction projections. Present as proposals to a city council with model-backed evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Urban Land Cover Composition
- **INTERNAL:** Surface Heat Absorption Rate, Waste Heat Generation, Evapotranspiration Cooling Rate, Ambient Air Temperature, Heat-Related Health Impact

Relationships

- **Urban Land Cover Composition to Surface Heat Absorption Rate:** POSITIVE (+) for impervious surface — Higher proportions of dark impervious surfaces (asphalt albedo 0.05-0.10, dark roofs 0.10-0.20) absorb 80-95% of incoming solar radiation, converting it to stored thermal energy. Each additional percentage of impervious cover increases the total heat absorbed across the urban area.
- **Urban Land Cover Composition to Evapotranspiration Cooling Rate:** NEGATIVE (-) for impervious surface — Replacing vegetation with impervious surfaces eliminates evapotranspiration — the most powerful natural cooling mechanism available in urban environments. Each tree removed eliminates the equivalent of 2-5 air conditioners of cooling capacity. The relationship is not linear: the first trees lost in a neighborhood have the greatest marginal cooling impact.
- **Surface Heat Absorption Rate to Ambient Air Temperature:** POSITIVE (+) — Higher heat absorption by urban surfaces raises air temperature through convective heat transfer (hot surfaces warm the air above them) and longwave radiation (surfaces re-emit stored heat as infrared radiation, especially at night). The nighttime effect is critical: high-thermal-mass materials release stored heat for hours after sunset.
- **Waste Heat Generation to Ambient Air Temperature:** POSITIVE (+) — Vehicles, air conditioning exhaust, industrial processes, and building energy use add anthropogenic heat directly to the urban atmosphere. In dense city cores, waste heat can contribute 2-4°F to ambient temperature. The air conditioning feedback loop is self-reinforcing: higher temperatures → more AC use → more waste heat → higher temperatures.
- **Ambient Air Temperature to Heat-Related Health Impact:** POSITIVE (+) — nonlinear above thresholds — Heat-related illness and mortality increase nonlinearly above critical temperature thresholds. Below ~95°F, most healthy people thermoregulate effectively. Above 105°F with high humidity, outdoor activity becomes dangerous for everyone. Above wet-bulb temperatures of 95°F, the human body cannot cool itself regardless of shade or hydration. Urban heat islands push cities across these thresholds for more hours and more days per year.

Key Discoveries

- Urban heat islands can amplify temperatures by 5-12°F during the day and up to 22°F at night compared to surrounding rural areas — the nighttime effect is especially dangerous because it prevents the human body from recovering from daytime heat stress
- Within a single city, temperature differences of 10-15°F exist between neighborhoods — and these differences map directly onto historic patterns of racial segregation and disinvestment, with formerly redlined neighborhoods having the least tree cover and the highest temperatures today
- Heat is the deadliest weather phenomenon in the United States, killing more people annually than hurricanes, floods, and tornadoes combined — and climate change is projected to increase heat-related deaths 3-5x by 2050 in cities that do not invest in cooling infrastructure
- A single mature tree provides cooling equivalent to 2-5 window air conditioning units through evapotranspiration, and increasing urban tree canopy from 10% to 30% can reduce neighborhood temperatures by 5-10°F — but tree planting takes 15-20 years to reach full cooling effect, requiring long-term planning

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Urban heat islands are just cities being a few degrees warmer — it's not a big deal"	A 'few degrees' of urban heat amplification becomes lethal when added to already-hot baseline temperatures and climate change. If a rural area reaches 98°F and the urban heat island adds 10°F, the city center reaches 108°F — well above dangerous heat thresholds. At night, when rural areas cool to 75°F, the city may only cool to 90°F, preventing physiological recovery from daytime heat stress. These temperature differences are measured in hospital admissions and deaths: heat-related mortality increases 2-5% for every 1°F above local heat thresholds. The 'few degrees' of urban heat represent the difference between uncomfortable and dangerous.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Just plant more trees everywhere — that solves the problem"	Tree planting is the most effective single cooling strategy, but it faces significant real-world limitations. Trees require 15-20 years to reach full canopy size and cooling capacity — they do not provide immediate relief during the current heat crisis. Many of the hottest urban areas have minimal unpaved space available for tree planting. Some urban soils are so compacted or contaminated that trees cannot establish without extensive remediation. Tree species selection must account for climate projections — trees planted today must survive in 2050's climate. And critically, tree planting without anti-displacement protections can trigger gentrification that displaces the vulnerable residents it was meant to protect. A comprehensive approach requires trees AND cool roofs AND shade structures AND community cooling centers AND equitable development policies.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Heat affects everyone equally — it's just weather"	Heat exposure is profoundly unequal within cities. Wealthy neighborhoods with mature tree canopies, parks, and well-insulated buildings can be 10-15°F cooler than low-income neighborhoods with minimal vegetation and aging housing stock. Vulnerability is compounded by social factors: outdoor workers (construction, agriculture, delivery) face unavoidable exposure; elderly people and young children thermoregulate less effectively; people without air conditioning have no escape during heat waves; and chronic health conditions (diabetes, cardiovascular disease, respiratory illness) make heat more dangerous. These vulnerability factors concentrate in the same neighborhoods that are physically hottest — creating a double burden where the people least able to cope with heat live in the places that produce the most of it.